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FDI inflows and export quality: Domestic competition and within-firm adjustment[☆]Qing Liu^a, Larry D. Qiu^b, Chaoqun Zhan^{c,*}^a School of Economics, Renmin University of China, China^b Department of Economics, Lingnan University, Hong Kong SAR, China^c Department of Accounting, Economics and Finance, Hong Kong Baptist University, Hong Kong SAR, China

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ABSTRACT

Product quality is widely regarded as an important determinant for economic development. This paper investigates whether horizontal foreign direct investment (FDI) improves or deteriorates the quality of domestic firms' exports. We use China's FDI regulation changes in 2002 as an instrument variable (IV) for FDI penetration in China to identify the causal impact and introduce a theoretical model to rationalize our empirical work. We find that FDI inflows exert a significantly *negative* effect on Chinese firms' export quality. The mechanism of the negative effect is that FDI intensifies the *domestic* market competition, which induces within-firm adjustment of product mix and lowers domestic firms' incentive to invest in the quality of new products. In particular, while domestic firms drop some existing products and introduce new products, they invest less in the quality of new products and maintain the quality of continuing products.

1. Introduction

How globalization affects innovation is an important question because innovation is arguably the most essential driver of economic growth. Improving the quality of existing products (i.e., quality upgrading) is found to be the most important form of innovation (Garcia-Macia et al., 2019). Quality is also quantitatively the most significant source of firm heterogeneity (Hottman et al., 2016). These latest research findings have brought the topic of product quality to the forefront of research. Foreign direct investment (FDI) has been regarded as an important vehicle for the diffusion of technology and know-how, which can potentially affect the quality upgrading of firms in the host country. In spite of the large amount of literature on the impacts of horizontal FDI on productivity, relatively limited work has been conducted on product quality.¹ Even less is known about the mechanisms through

which FDI affects firms' quality and how firms make quality adjustments across products. In this study, we investigate whether and how FDI affects the export quality of firms in the host country.

Our analysis focuses on China, which has three advantages. First, China relaxed the FDI regulations upon its accession into the World Trade Organization (WTO), which provides us a quasi-natural experiment to identify the causal effects of FDI on the export quality of Chinese firms (Lu et al., 2017; Eppinger and Ma, 2024).² To comply with the commitments in many rounds of negotiations, China implemented an FDI policy reform in 2002 after its accession into the WTO in 2001. The reform relaxed restrictions on FDI entry in approximately one-fourth of manufacturing industries, thereby causing exogenous variations in FDI entry across industries. We follow the literature to employ this reform as an instrument variable (IV) for industrial FDI

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¹ There are some exceptions, including Swenson and Chen (2014), Harding and Javorcik (2012) and Bajgar and Jorvorcik (2020). Although they all examine export quality in general, their focuses are different from ours. For example, Harding and Javorcik (2012) focus on industry-level export quality. We will discuss them in detail in the literature part.

² Throughout this study, we define Chinese firms as indigenous Chinese firms without foreign equity. We are not able to examine the effects on products sold to the domestic market because of the lack of product-level data of the Chinese firms in the domestic market. However, our model and analysis imply that the same results found using the export products should also apply to the domestic products.

intensity. Second, detailed data of Chinese firms are available, which enables us to investigate the FDI effect and explore the mechanisms. Specifically, our analysis is based on an annual panel during 2000–2006 of Chinese firms with balance sheet data matched with detailed customs data, at the levels of firm, six-digit Harmonized System (HS) product, destination, and year. Third, China is one of the largest FDI recipients and the largest exporter of manufacturing products in the world. Thus, using Chinese data to examine the impact of FDI on export quality is of great importance.

Empirically, we identify a robust negative effect of FDI on Chinese firms' export quality during 2000–2006. This finding is in sharp contrast to the existing literature for other developing countries (e.g., [Harding and Javorcik, 2012](#)). Specifically, we find that Chinese firms in industries with higher FDI intensity tend to have on average lower export quality. The result is obtained using a quality measure proposed by [Khandelwal et al. \(2013\)](#), but also holds when using unit values or relative unit values. The effect is not only statistically significant but also economically important. We find that a one-percentage-point increase in FDI intensity decreases export quality by 1.77%. Since FDI intensity has on average increased 6.12 percentage points from 2000 to 2006; thus, the resulted reduction in export quality is 10.83%.

We then build a single-product model to help understand our main finding and unveil the mechanism. A well-known insight in the literature is that horizontal FDI generates not only a positive effect on local firms of the host country through technology spillover, but also a negative effect by increasing market competition (also called market stealing as in [Aitken and Harrison, 1999](#)). In our model, we incorporate these two effects and assume that a typical Chinese firm makes an investment to determine the quality of its product sold to both the domestic and international markets. We show that the firm's product quality increases with the spillover effect, but decreases with the competition effect. Under certain conditions, the negative competition effect dominates the positive spillover effect, and thus, the Chinese firms export on average lower-quality products when there is more FDI entry. This analysis explains the aforementioned empirical finding.

We provide direct empirical evidence to support the insight of the model, that is, the FDI effect works through intensifying competition in the domestic market. In particular, we find that after 2002, firms in the reformed industries significantly reduced their domestic sales, but not the export sales, compared with firms in the unreformed industries. In addition, FDI's negative effect on export quality is more pronounced for firms that initially rely more on the domestic market. Our placebo test on pure exporters shows no significant effect of FDI on their export quality.

To investigate how firms adjust their export quality across products within firms, we extend our single-product model to a multiproduct setting, similar to [Bernard et al. \(2010\)](#). In particular, each firm draws a capability of multiproduct management, and the productivity of all its products. There exist various shocks of consumer taste on all products. A firm makes an investment on the quality of each of its products. While our analysis predicts a number of outcomes in response to FDI liberalization, the most interesting finding is that the quality of the newly introduced products is lower than that of the products existing before the FDI liberalization.

We find empirical evidence consistent with the predictions of our multi-product model. First, the average quality of continuing products remains unchanged between the treated and control groups after the reform. Second, the average quality of new products is not significantly different from that of continuing products in the pre-reform period but is significantly lower in the post-reform period in the treated group (i.e., the reformed industries), compared with the counterpart in the control group (i.e., the uninformed industries). Third, the average quality of the dropped products is the same as that of continuing products both in the pre- and post-reform periods. Fourth, the product scope is not significantly affected by the FDI entry, thereby implying that firms add new products to replace dropped products. Thus, we

can conclude that FDI reduces the average quality of a Chinese firm's export products through reducing the quality of the new products. This FDI-induced reduction in the quality of new products could profoundly influence the firm's and the economy's growth.³

Our paper makes contributions to the large amount of empirical literature that studies the impacts of FDI on domestic firms' performance. Although the majority of studies in this literature focus on the productivity of the domestic firms, a few studies focus on the product quality of the domestic firms. Our paper belongs to the latter group.

FDI can affect domestic firms' productivity through selection and market reallocation (e.g., [Alfaro and Chen, 2018](#)), vertical linkages (e.g., [Javorcik, 2004](#)), and horizontal spillover (e.g., [Aitken and Harrison, 1999](#)). Most of the studies in this group focus on horizontal FDI spillover effects, but the findings are mixed ([Aitken and Harrison, 1999](#); [Haskel et al., 2007](#); [Keller and Yeaple, 2009](#); [Javorcik and Spatareanu, 2008](#); [Lu et al., 2017](#)). As for China, [Lu et al. \(2017\)](#) find negative spillover effects of FDI, suggesting that the negative competition effect is larger than the positive technology spillover on Chinese firms' total factor productivity (TFP) and other performances. However, [Lu et al. \(2017\)](#) do not consider export product quality, which is the focus of the present study.

Our paper is mostly related to three existing studies on export quality. The first study is that of [Harding and Javorcik \(2012\)](#), which examines the effects of horizontal FDI on export upgrading. [Harding and Javorcik \(2012\)](#) find a positive effect for a sample of developing countries, but we find a *negative* effect of FDI on Chinese firms' export quality. Our explanations for these opposite findings are as follows. First, the positive effect of [Harding and Javorcik \(2012\)](#) is the average effect of developing countries in their sample from 1984 to 2000. Our focus is on China – the largest developing country, the largest FDI recipient among developing countries, and the largest exporter of manufacturing in the world. Thus, the negative effect we find is of particular importance and interest. Second, [Harding and Javorcik \(2012\)](#) use trade statistics that do not distinguish between exports by domestic and foreign firms, and thus, their analysis captures the sum of the effects on the two types of firms. FDI firms typically produce higher-quality products in developing countries than the local firms. Therefore, if FDI firms use the developing countries as their export platforms, then the export quality of the developing countries will naturally be higher after the FDI inflows than before. This finding implies that if the share of FDI firms' exports is sufficiently large, then the positive effect of [Harding and Javorcik \(2012\)](#) may still exist even if the true effect on domestic firms' export unit values is negative. In contrast, our data allows us to separate the FDI firms, also called foreign-invested enterprises (FIEs, which include both the wholly foreign-owned firms and Sino-foreign joint ventures) in China, and the indigenous domestic firms. We find a negative effect of FDI on domestic firms' export quality. Third, [Harding and Javorcik \(2012\)](#) use export unit values to represent export quality, whereas we use a more sophisticated measure of product quality proposed by [Khandelwal et al. \(2013\)](#). We also use export unit values as an alternative measure in the robustness analysis and find that our result is robust. Fourth, we use firm-product data, whereas [Harding and Javorcik \(2012\)](#) do not have the firm information. Thus, the quality adjustment in our finding is within firms, whereas that in [Harding and Javorcik \(2012\)](#) includes cross-firms changes.

³ The quality of the exported product is often regarded as the quality frontier of the economy. Our results indicate that the horizontal FDI penetration slows down the upgrading of the quality frontier, and therefore the growth of the host economy. In addition, how a firm adds new products matters for its growth. Introducing *innovative* new products is the key to firm growth, as the newly acquired technical competence can be applied to the horizontal expansion of the firm's product mix ([Braguinsky et al., 2021](#)). Lower-quality products usually are less innovative. Thus, FDI's negative impact on the quality of *new* products suffocates the spillover effect from introducing innovative new products, which further impedes the firms' growth.

The second study that our paper is closely related to is the working paper of Swenson and Chen (2014), which examines the FDI effect on the quality of new export products from China. Once again, our finding and that of Swenson and Chen (2014) are opposite, which is interesting because both studies address more or less the same issue from China, albeit with different time coverages. Thus, understanding the reasons is important. First, we use an IV approach, based on industry variations, to identify the causal effects of FDI on export quality. The result of Swenson and Chen (2014), based on regional variations, is about a correlation between FDI and export quality. Second, we measure the presence of FDI by using the output share of the FIEs in each industry, whereas Swenson and Chen (2014) use the export volume of the FIEs. These two measures represent different aspects/types of FDI and have different implications on the mechanisms through which FDI exerts impacts on export quality. Our measure captures the penetration of FIEs in each industry, and thus, FDI affects the Chinese firms mainly through changing competition and technology spillover in each industry. In contrast, the measure of Swenson and Chen (2014) captures the export activity of FIEs, and therefore, their analysis is more about Chinese firms' learning from FIEs' export.⁴

Finally, using Romanian data, Bajgar and Jorvorcik (2020) find a positive relationship between the presence of multinationals (MNEs) in the upstream industries and the export quality of the Romanian firms in the downstream industries. Such a relationship is also found for the presence of MNEs in the downstream and the same industries, but is less robust. Although our focus is on horizontal FDI (MNEs and domestic firms in the same industry), we also include upstream and downstream FDI in our robustness analysis. On the one hand, we include these types of FDI in the analysis to control for more variables that may affect Chinese firms' export quality. Our result on the effect of horizontal FDI on export quality is robust to the inclusion of upstream and downstream FDI. On the other hand, we also find positive effects of upstream and downstream FDI on Chinese firms' export quality, although the statistical significance of the effects is not robust.⁵

Our study also contributes to the literature on the effects of trade liberalization on export quality. While existing literature, such as Bas and Strauss-Kahn (2015) and Fan et al. (2015), discovers the benefits of enhanced input accessibility due to trade liberalization, our paper presents a different result of FDI liberalization. Specifically, FDI introduces both positive technology spillover effects and negative competitive pressures. We find that FDI exerts an overall negative effect on Chinese firms' export quality.

This paper is organized as follows. Section 2 describes China's FDI policy and data. Section 3 introduces the econometric specification of the our empirical model and presents the main findings. Section 4 checks the robustness of the main findings. Section 5 introduces a theoretical model to explain the main findings and suggests the mechanism behind the FDI effects. Section 6 provides further empirical evidence to support the mechanism. Section 7 concludes the paper.

2. Institutional background and data

2.1. Institutional background

China's historical approach to FDI was characterized by a closed-door policy until 1978 when the nation embarked on the "Reforming and Opening-up" initiative. Despite this initial shift towards openness,

FDI entry remained heavily regulated, with significant barriers in place, before 2002 when China entered the WTO.

The process of opening up accelerated from 1995 onwards. China's efforts to integrate into the global trade system were evident with its bid to rejoin the General Agreement on Tariffs and Trade (GATT) in July 1986, followed by the official submission of its WTO accession application in 1995. In 1995, the National Development and Reform Commission and the Ministry of Foreign Trade and Economic Cooperation jointly published the inaugural "Catalogue for the Guidance of Foreign Investment Industries". The Catalogue, which categorized products into various classifications based on their openness to foreign investment, served as the government guidelines for regulating FDI inflows.

The negotiations for China's WTO accession were protracted and challenging, spanning a duration of 15 years. In order to comply with the Agreement on Trade-Related Investment Measures, China committed to eliminating restrictions and distortions on foreign investment and international trade. Moreover, China accepted additional obligations that substantially exceed the standard WTO rules during the negotiation (Qin, 2007).

A significant milestone in China's FDI liberalization occurred in 2002 with the revision of the Catalogue, which was a direct consequence of China's accession to the WTO in 2001 (Eppinger and Ma, forthcoming). Notably, more than one-fourth of the manufacturing industries experienced relaxation in FDI entry under this liberalization. Consequently, the relaxed industries witnessed increased FDI inflows. In this study, we use this arguably exogenous FDI liberalization in 2002 to identify the FDI's effect on export quality.

2.2. Data

We combine three datasets to analyze the effects of FDI on indigenous domestic firms' export product quality. The datasets are the industry-level FDI policy data, firm-level balance sheet data, and transaction-level trade data.

□ Industry-level FDI Policy Data

FDI entry was highly regulated in China mainly through the Catalogue for the Guidance of Foreign Investment Industries (henceforth, the Catalogue). In June 1995, the Chinese government promulgated the Catalogue, which explicitly classifies all industries/products into four categories, depending on whether FDI is (i) supported, (ii) permitted, (iii) restricted, or (iv) prohibited. Since the promulgation, the Catalogue has been revised several times. The first revision was made in 1997. The second revision was announced in March 2002 soon after China's accession to the WTO in December 2001 and was substantial, with more than one-fourth of the manufacturing industries experienced relaxation in FDI entry under this reform. This revision was mainly the result of China's commitment in WTO accession to reduce FDI entry restrictions in a wide range of industries and to fully comply with the "Agreement on Trade-Related Investment Measures" for the country's WTO accession (Lu et al., 2017). As shown by the literature, due to the politically complicated negotiation process of China's WTO accession, this regulation reform a plausibly exogenous shock to China's manufacturing industries (Lu et al., 2017; Eppinger and Ma, 2024). Subsequent revisions were made in 2004, 2007, and 2011, respectively, to further relax restrictions on FDI entry.

In this study, our data sample period is 2000–2006. Accordingly, we use the revision of the Catalogue in 2002 as the exogenous shock to FDI inflows to identify the effect of FDI on indigenous domestic firms' export quality. The 2002's revision also provides substantial variations for empirical identification as more than 25% of the manufacturing industries experienced relaxation in FDI entry under this reform. Although 2004 falls within the data sample period, we do not use the 2004's revision of the Catalogue because it involves only a few minor modifications. We do not use the versions of the Catalogue in 2007 and 2011 because they are out of the sample period.

⁴ Other studies have been conducted on the quality of export from China not related to FDI. For example, Manova and Yu (2017) find that the quality of newly exported products from China is lower than that of the incumbent ones. Rodrik (2006), Schott (2008), and Wang et al. (2010) document and explain the exceptionally high sophistication of Chinese export products.

⁵ Javorcik et al. (2018) also explore the effect of vertical FDI, but they focus on production complexity instead of quality. They find a robust positive effect of downstream FDI entry on domestic firms' production complexity in Turkey.

We construct the industry-level policy data by comparing the 1997 and 2002 versions of the *Catalogue*. We construct data at four-digit Chinese Industry Classification (CIC) level. Following Lu et al. (2017), we focus on whether there are changes for all products regarding FDI entry regulation in the 2002 *Catalogue* compared to the 1997 *Catalogue*. We first classify all products into three groups according to the changes, and they are “more welcome”, “more restrictive”, and “no change”. We then define FDI policy change for each four-digit CIC industry which belongs to one of the following four categories: (i) FDI relaxed industry, in which no products is “more restrictive” and at least one product is “more welcome”; (ii) no-change industry, in which all products experience no change in FDI regulation in 2002; (iii) FDI restricted industry, in which no product is “more welcome” and at least one product is “more restrictive”; and (iv) mixed industry, in which some products are “more welcome” but some are “more restrictive”. Based on the above classification, the last two categories together contain only 15 industries in our data. Moreover, our objective is to examine the effects of increased FDI inflows. Thus, we exclude the last two categories from our data set. As a result, we obtain 126 FDI relaxed industries, which will be used as our treatment group in our empirical analysis, and 3 no-change industries, which will be used as our control group. For convenience, we refer to these two groups of industries as the *reformed industries* and *unreformed industries*, respectively.

□ Firm-level Balance Sheet Data

Second, the firm-level balance sheet data. We build this data set based on the Above-Scale Industrial Firms Panel (ASIFP) database maintained by China's National Bureau of Statistics (NBS). The ASIFP database includes all state-owned enterprises (SOEs) and non-SOEs with annual sales of at least 5 million RMB in mining, manufacturing, and utilities industries. The number of firms in this database varies over time from over 140,000 in the late 1990s to over 336,000 in 2007. The database contains detailed information of each firm, including the firm's official name, industry, location, ownership type, age, employment, capital, intermediate inputs, and new product sales. We clean the ASIFP database as follows. We exclude non-manufacturing firms. We drop firms according to the basic rules of Generally Accepted Accounting Principles (e.g., firms with the sum of their liquid assets, total fixed assets, or net value of fixed assets greater than total assets). We also drop firms with fewer than eight workers because those firms are subject to a different legal regime and firms with obviously wrong information (e.g., establishment year is 1900 or earlier). Moreover, we exclude firms with positive foreign capital share to focus on indigenous domestic firms.⁶ Our sample period is 2000–2006, mainly due to the data availability of the Custom data.

□ Transaction-level Trade Data

The Chinese Customs maintain all export transactions at firm-destination-product-year level. The data of each transaction include information of the eight-digit Harmonized System (HS) product code, free-on-board value, quantity, source city, destination country, trade type (e.g. ordinary trade or processing trade), contact information of the firm, and others. We aggregate data into six-digit HS product level based on the HS 2002 classification using the concordance provided by UN Comtrade.

We adopt several filters to clean the data so that it is suitable for our study. First, we exclude trade intermediaries because our focus is on manufacturing firms' production and export decisions. We follow Ahn et al. (2011) to identify trade intermediaries through keywords in their names. Second, we exclude processing trade transactions because

we are interested in how indigenous domestic firms adjust the export quality of their own products in response to the change in FDI inflows. It is not difficult to see that the quality of processing trade products (e.g., iPhone assembled in China), which is set by the foreign partners, is mainly affected by the conditions in the foreign markets where the products are sold. Third, to examine the effects of FDI policy change in 2002 on export product quality, we need information about the firm's export product before and after the policy change. Accordingly, our data set only keeps firms that had exports before 2002 and also had exports in at least one year in 2002–2006.

Finally, we merge the above three data sets. We first merge the ASIFP and the FDI policy data sets according to the four-digit CIC industry classification, which will then be merged with the Customs data set using the common contact information of the firms in the ASIFP and Customs data (e.g., company name, telephone number, zip code and contact person).⁷ As a result, we obtain an unbalanced panel data with transaction-level data of all indigenous domestic exporters during 2000–2006 with around 0.4 million observations.

Our working sample consists of a subset of exporters drawn from the matched sample of the ASIFP and Custom data. These exporters are slightly larger than those exporters in the matched sample in terms of sales, value of exports, and employment. The former are much larger in these measures than the firms in the ASIFP sample.⁸ This pattern is consistent with the literature using similar datasets, e.g., Yu (2015).

3. Empirical strategy and baseline results

3.1. Empirical strategy

We introduce the following benchmark model to examine the effects of inward FDI on Chinese firms' export quality:

$$Quality_{fihct} = \beta FDI_Intensity_{it} + \mathbf{X}'_{fit}\gamma + \lambda_f + \lambda_h + \lambda_{ct} + \epsilon_{fihct}. \quad (1)$$

In the above model, $Quality_{fihct}$ is a variable to capture the quality of firm f 's export of product h to country c in year t , where firm f belongs to industry i . Products are defined at the HS six-digit level and industries are defined at the CIC four-digit level of the 2003 version. $FDI_Intensity_{it}$ is FDI intensity of industry i in year t to capture the presence of FDI. $\mathbf{X}_{fit}$ is a vector of time-varying firm and industry characteristics that might confound the effect of FDI on export quality. The time-varying firm controls include firm total export, total sales, total factor productivity (OP method), capital-labor ratio, and a State Owned Enterprises (SOE) indicator. The time-varying industry covariates include variables related to FDI policy determinants and tariff reductions because of China's WTO accession, which will be explained in detail later in this section.

Variables λ_f , λ_h , and λ_{ct} are firm, product, and destination-year fixed effects, respectively. The firm fixed effect λ_f allows us to control for firm-specific characteristics that may affect the quality of the product. Product fixed effects λ_h is introduced to make quality more comparable across products. The destination-year fixed effect λ_{ct} controls for time-varying characteristics of the destinations that affect firm's quality estimate, such as the average income level and aggregate price level in the destination market. Although the FDI intensity varies only at the industry-year level, the quality measure may vary across products and destinations. Therefore, we cluster error terms ϵ_{fihct} at the industry-year level to allow them to correlate within the same industry and year.

There exist many ways of measuring product quality. We mainly adopt the approach developed by Khandelwal et al. (2013). The assumption/justification of this approach is that conditional on the same

⁶ If a firm has foreign capital, the firm's business decisions may be influenced by the foreign firms. Thus, the FDI's effects on the firm's export product quality are likely to be different from those on pure domestic firms. However, we can also keep firms with a small share of foreign capital, say 25%, in our data set. We find that our main results do not change with those modifications of the data construction.

⁷ In such a merge exercise, 94% are matched by company names, 4% are matched by zip code and contact person, and the remaining 2% are matched by telephone number.

⁸ See details in Table A.1 in Appendix.

price, products with higher market share must have higher quality. Accordingly, we can estimate the quality of each firm-destination-product as the residual from the following OLS regression:

$$\ln x_{fihct} + \sigma \ln p_{fihct} = \lambda_h + \lambda_{ct} + v_{fihct},$$

where x_{fihct} is the quantity of the product, p_{fihct} is the price of the product, σ is the elasticity of substitution between products in the market, λ_{ct} is the destination-year fixed effect, which captures the country's total income and price index, and λ_h is the product fixed effect, which controls for time-invariant product characteristics. The estimated quality is $\ln q_{fihct} = \hat{v}_{fihct}/(\sigma - 1)$. Using the estimated elasticity for China at the HS three-digit industry from Broda et al. (2017), we run the above regression for each HS three-digit industry to calculate our baseline measure for quality ($\ln q$).⁹

For robustness check in a later section, we will also construct alternative quality measures. Khandelwal et al. (2013) and Fan et al. (2015) calculate quality using the same framework as above but assuming a specific elasticity for all industries. Following Fan et al. (2015), we use specific values $\sigma = 5$ or $\sigma = 10$ to calculate two alternative quality measures ($\ln q5$ and $\ln q10$). In addition, we measure quality using log unit price of exports ($\ln p$) and demeaned log price of exports ($r \ln p$), which have been adopted in the literature.

We use the standard measure from the FDI-spillover literature to measure FDI intensity as the following:¹⁰

$$FDI_Intensity_{it} = \frac{\sum_{f \in \Omega_{it}} FDI_{fit} \times Output_{fit}}{\sum_{f \in \Omega_{it}} Output_{fit}}, \quad (2)$$

where Ω_{it} is the set of all firms of industry i in year t , FDI_{fit} is the foreign equity share in firm f , and $Output_{fit}$ is firm f 's output.

There may be many concerns about our baseline specification (1), but the most serious one must be about endogeneity. Specifically, the identification of β depends on the assumption that $FDI_Intensity_{it}$ is uncorrelated with the error term ϵ_{fihct} conditional on all control variables. This assumption may not hold in general because FDI decisions are made by MNEs and likely to be affected by factors that also affect the Chinese firms' product quality. To deal with this endogeneity problem, we utilize Chinese FDI policy change in 2002 to identify the causal relationship between FDI and export quality. The underlying rationale is that the differential relaxation of FDI regulations will generate exogenous variations of FDI intensity across industries, which are uncorrelated with firms' export quality. Hence, the policy reform can be used as an IV to identify the causal effects.

We implement the above idea as follows. First, we introduce the industry-level FDI reform indicator, denoted by FDI_Lib_i , with $FDI_Lib_i = 1$ if industry i is a reformed industry, i.e., FDI is more encouraged in the 2002 Catalogue than in the 1997 Catalogue, and $FDI_Lib_i = 0$ if the industry is an unreformed one. Second, we introduce the reform period indicator, denoted by $POST02_t$, with $POST02_t = 1$ if $t \geq 2002$, and $POST02_t = 0$ if $year < 2002$.¹¹ Finally, we construct the IV for $FDI_Intensity_{it}$, which is $FDI_Lib_i \times POST02_t$.

3.2. Validity of the identification

The validity of our IV strategy relies on two conditions. First, the Relevance Condition requires the instrument to be highly correlated with the key explanatory variable $FDI_Intensity_{it}$. Second, the

Orthogonal Condition, which requires the instrument to be conditionally uncorrelated with the error term in the first stage, that is, $Corr(FDI_Lib_i \times POST02_t, \epsilon_{fihct} | X'_{fit}, \lambda_f, \lambda_h, \lambda_{ct}) = 0$. We will verify the Relevance Condition later based on our estimation of the correlation. As for the Orthogonal Condition, it cannot be tested directly, and thus, we discuss it now from three perspectives.

The first perspective is the selection of the industries to reform. One possibility of violating the Orthogonal Condition is that the reformed industries are not selected independently from the industry's characteristics which also affect product quality. We have two responses to this concern. On the one hand, the selection of the reformed industries is not a decision made solely by the Chinese government; rather, it is the outcome of the negotiations between China and the WTO members for China's accession to the WTO. The negotiations on FDI entry targeted factors can hardly affect product quality. The industry average quality in the pre-reform period is statistically uncorrelated with the choice of the reformed industries, as we will show in a flexible estimation. On the other hand, even if we believe there are specific criteria that the Chinese government used to select the reformed industries, no official documents are available for us to analyze. However, we can still address this latter issue to a certain extent. Lu et al. (2017) characterized potential determinants of the changes in FDI regulations upon China's accession into WTO, which include the new product intensity, export intensity, number of firms, and the average age of the firms at the CIC four-digit level in the pre-reform period. We follow them to include interactions of these variables in the initial year (2000) Z_{i2000} with year dummies λ_t to control flexibly for quality changes that are caused by the endogenous selection of reformed industries. With the inclusion of those additional variables, we are more confident that FDI_Lib_i is conditionally uncorrelated with the error term.

The second perspective is the contemporary reforms. If there exist other policy reforms that were also introduced around 2002 and had differential effects on our treatment and control groups, then our baseline estimates would be confounded by the effects of those policy reforms. Regarding this concern, two policy changes that took place in the early 2000s should be considered, which are the SOE reform and China's accession into the WTO.

The SOE reform indeed started in the 1980s, but accelerated during the 2000s. The reform in the early 2000s resulted in large-scale privatization, close-down of small SOEs, and efficiency improvement of the surviving SOEs. Industries with *ex ante* higher share of SOEs tend to be more severely affected. Therefore, we add the interaction terms between year dummies and industry SOE share at the 2001 levels into our regression to control for potential confounding effects from SOE reforms.

In addition to FDI liberalization, China also experienced trade liberalization during the same period due to its WTO accession. China lowered its import tariffs significantly in almost all industries. As established in the literature that trade liberalization has a significant impact on export quality of firms, especially the impacts of input tariff reduction as examined by Fan et al. (2015), we also control explicitly for the tariffs on input and output imports of China. Moreover, although the foreign tariffs faced by China were generally low before China's accession into WTO and did not have big changes, we also control for the weighted average foreign tariffs faced by Chinese exports in our regression.¹²

The third perspective is the comparability between the reformed and unreformed industries in the pre-reform period. The estimate β from our baseline regression (1) is the average treatment effect of FDI reform on export quality, which is the difference between the treated and control groups in their average difference in export quality between

⁹ As this quality measure is orthogonal to product and country-year fixed effects by construction, we do not include these fixed effects in regressions when using it as the dependent variable.

¹⁰ This measure has been widely used in literature with different weights (e.g., output, employment, capital, and asset) to investigate FDI spillover effect. See Barrios et al. (2011) for discussions.

¹¹ The 2002 Catalogue was enacted on April 1, 2002. If we define the reform period indicator by $POST02_t = 1$ for $t \geq 2003$, our results will not change.

¹² The reformed and unreformed industries actually experienced similar levels of changes in import tariffs, foreign tariffs, and SOE shares after the reform.

Table 1
Flexible estimation.

Dependent variable: $\ln q$	(1)
$FDI_lib_i \times 2001$	0.0262 (0.0295)
$FDI_lib_i \times 2002$	-0.0303 (0.0291)
$FDI_lib_i \times 2003$	-0.0604** (0.0302)
$FDI_lib_i \times 2004$	-0.0655** (0.0314)
$FDI_lib_i \times 2005$	-0.0798** (0.0315)
$FDI_lib_i \times 2006$	-0.1223*** (0.0352)
Year FE	Yes
Firm FE	Yes
Controls	Yes
N	409 169

Note: We use the initial year, 2000, as the base year, to see whether the coefficients of the treatment years are different with that of non-treatment year 2001. Controls include all control variables, the same as those in column (5) of Table 2. Robust standard errors in parentheses, clustered at CIC four-digit industry-year level.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

the pre- and post-reform periods. The validity of this estimation depends on the comparability of the trends between the reformed and unreformed industries. To test this comparability assumption, we can conduct a flexible estimation by replacing the *POST02* dummy with a vector of year dummies λ_t , indicating the years from 2001 to 2006 with the initial year 2000 as the omitted reference year. This new specification imposes a less restrictive timing structure on the model. It allows us to examine how the differences in export quality between the reformed and unreformed industries vary over time. The estimation results will help answer two questions: whether the two groups have comparable trends before the reform and whether a divergence exists between the trends after the reform. Table 1 reports the results. We find that the coefficients in year 2001 and 2002 are insignificant, indicating that common pre-trends exist between the treatment and control groups. From 2003 to 2006, firms in the reformed industries have significantly lower quality, indicating a divergence from the treatment year.

These results provide support to our difference-in-differences (DID) specification.¹³

3.3. Graphical evidence

The impact of FDI on domestic firms' product quality could be checked roughly in a graph. Fig. 1 presents the distributions of export quality of Chinese firms in 2000 and 2006, respectively, with the left and right panels referring to firms in the reformed and unreformed industries, respectively. We remove the outliers, which are defined as those observations in the bottom and top 2.5th percentiles, to ensure that the curves represent the main features of the distributions. Based on visual observation, we notice that in the reformed industries, the distribution of export quality shifts to the left from 2000 to 2006, indicating that the average quality becomes lower, whereas we do not discern such a shift in the unreformed industries. This raw evidence suggests a hypothesis that the FDI policy change, which results in more FDI, reduces Chinese firms' export quality. We then check this hypothesis econometrically.

¹³ The common pre-trends assumption has also been identified using the pre-reform subsample. The reform and un-reformed industries have no significant differences in quality, quality growth, and many other important characteristics in the pre-reform period. In particular, we do not see significant differences in the number of firms, competition level, SOE share, FIE share between the two groups. These suggest that the lobby forces from large firms, SOEs, or multinationals may not be a critical concern in our context.

3.4. Baseline results

We first conduct the IV analysis. As our focus is on the effects of FDI on Chinese firms' export quality, our empirical analysis is based on the subset of our merged data which contains domestic firms only. In the first stage, we examine the effect of FDI policy change ($FDI_lib_i \times POST02_t$) on FDI intensity ($FDI_Intensity_{it}$). We report the IV estimation results in Table 2. Panel A presents the first-stage results, which test whether the FDI policy change affects FDI intensity. Panel B displays the second-stage results, which identify the causal effects of FDI intensity on export quality. In addition, we also report the reduced-form estimation results based on model (3) in Panel C.

$$Quality_{fihct} = \delta FDI_lib_i \times POST02_t + X'_{fict} \gamma + \lambda_f + \lambda_h + \lambda_{ct} + \epsilon_{fihct}. \quad (3)$$

In Table 2, we first include only firm and year fixed effects in column (1) to show the basic correlations. Focusing on column (1), Panel A shows that industries experienced relaxation in FDI regulation have, on average, higher FDI intensity in the post-reform years than those in the unreformed industries by 2.18%. The IV estimate in panel B identifies a significantly negative relationship (-0.9833) between the FDI intensity of an industry and the export quality of the firms in that industry. This IV analysis allows us to establish our main result, that is, *inward FDI reduces Chinese firms' export quality*. The reduced-form estimation in Panel C (-0.0414) confirms this result.

We introduce additional controls to address the concerns discussed in the preceding subsection. In column (2), we add a set of interactions of year dummies with pre-reform industrial characteristics to deal with the concern of nonrandom selection of reform industries following Lu et al. (2017). In column (3), we include an interaction between the share of SOE firms in each industry in 2001 and year dummies to control for the ongoing SOE reforms.

In column (4), we control for the average import output tariffs, import input tariffs, and foreign tariffs faced by Chinese exports to tease out the effect of trade liberalization. In particular, we report the coefficients of import output and input tariffs. The results show that the reduction of import output tariffs leads to a decrease in export quality, consistent with the intensified competition brought by imports. In addition, our results on import input tariffs are also consistent with the finding of Fan et al. (2015) that there exists a positive effect of import input tariff reduction on the export quality of Chinese firms, especially in the reduced form regressions. Notably, our results indicate that the impact of FDI remains significantly negative and unchanged in magnitude. This suggests that the influence of FDI on export quality is different from the effects of input tariffs, highlighting a novel dimension through which globalization affects firm-level export quality.

Although we have controlled a battery of industry-level characteristics, to avoid the bias due to omitted variables, we further include a vector of time-varying firm-level covariates in column (5). Specifically, we include the interaction terms between the initial firm size and capital-labor ratio with year dummies.¹⁴ The coefficients on *FDI* and $FDI_lib_i \times POST02_t$ are significantly negative and highly robust in their magnitudes compared to previous columns with industry-level controls only. In sum, our main finding is robust to the omitted firm-level variable concern. These results confirm that the negative effect of FDI on export quality is not driven by contemporary reforms, potential selection bias, and firm performances and characteristics.

The estimates are not only statistically significant but also economically important. Based on the second-stage IV estimate in column (5), a one-percentage point increase in FDI intensity reduces 1.55 percent of the measured quality. Since FDI intensity has on average increased 6.12 percentage points from 2000 to 2006, the increase in FDI reduces the measured export quality in 2006 by 9.49% compared to the level in 2000.

¹⁴ We do not include the time-varying firm-level outcomes to alleviate the potential bad control problem. Nevertheless, our results remain robust if doing so.

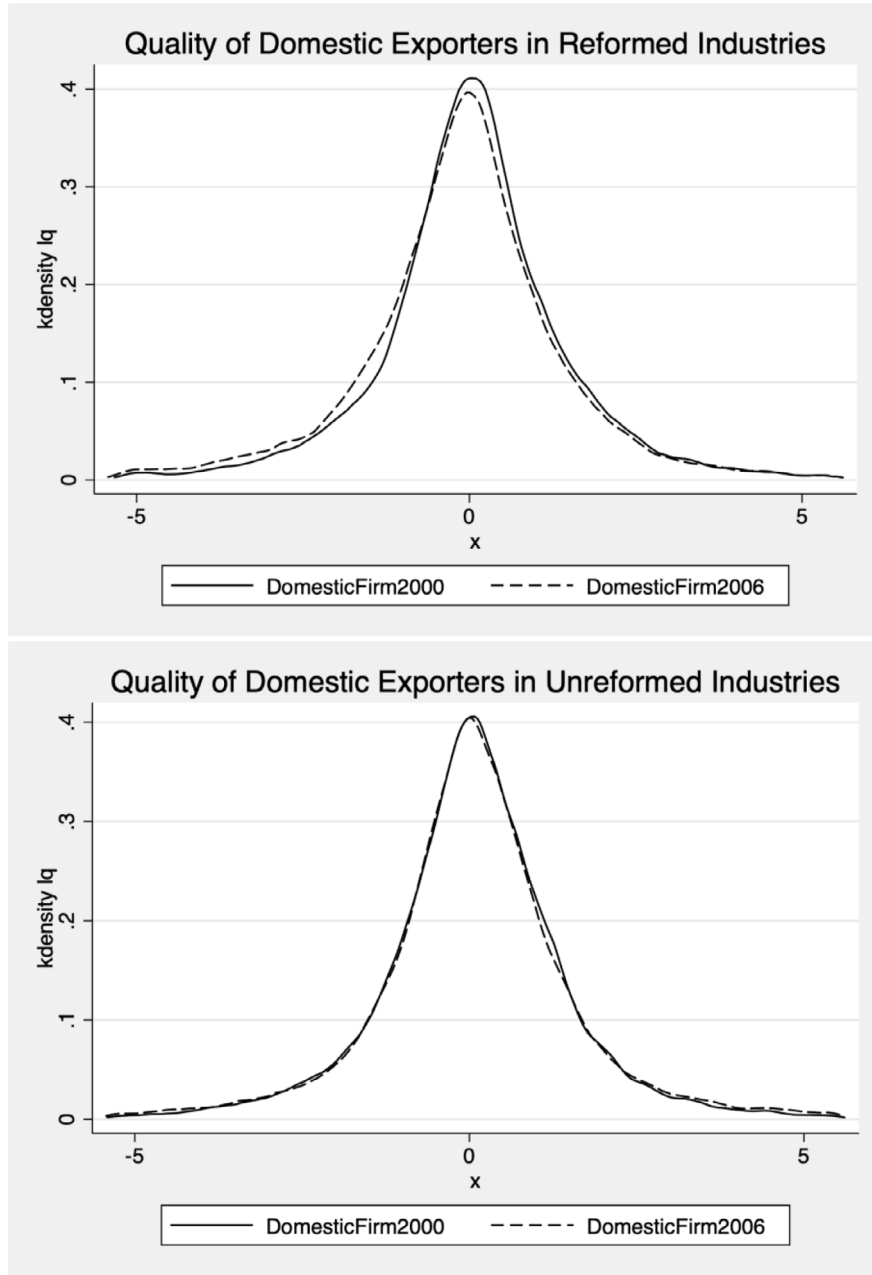


Fig. 1. Quality distribution of domestic exporters in FDI reformed and unreformed industries in 2000 and 2006.

4. Robustness checks

4.1. Upstream and downstream FDI

While our focus is the impact of horizontal FDI on Chinese firms' export quality, vertical FDI may also have influences (Bajgar and Jorvorcik, 2020). For example, foreign affiliates can have a positive impact on their local suppliers in China due to the former's demand for high quality inputs. Foreign affiliates can also benefit their local downstream buyers in China by supplying high quality inputs. When the FDI regulation is relaxed, there will be more horizontal FDI as well as vertical FDI, both upstream and downstream. Therefore, our baseline finding may be confounded by the vertical FDI.

To deal with this concern, we first follow Javorcik (2004) to construct the upstream and downstream FDI intensity indexes for each

industry i in year t as follows:

$$FDI_Intensity_Up_{it} = \sum_{j \neq i} a_{ji} \times FDI_Intensity_{jt},$$

$$FDI_Intensity_Down_{it} = \sum_{j \neq i} a_{ij} \times FDI_Intensity_{jt},$$

where a_{ij} is the share of industry i 's output (as the upstream) used in industry j 's production (as the downstream) and $FDI_Intensity_{jt}$ is the horizontal FDI intensity in industry j as defined in (2). We calculate a_{ij} based on Chinese Input-output Table of the 2002 version. Following the IV approach adopted earlier for horizontal FDI, we use the average extent of FDI policy relaxation weighted by input-output linkages to instrument for the upstream and downstream FDI intensity. Specifically, the instruments for $FDI_Intensity_Up_{it}$ and $FDI_Intensity_Down_{it}$ are $(\sum_{j, j \neq i} a_{ji} \times FDI_lib_j) \times POST02_t$ and $(\sum_{j, j \neq i} a_{ij} \times FDI_lib_j) \times POST02_t$, respectively.

We run regressions by adding the upstream and downstream FDI intensity to the baseline model and report the results in column (1)

Table 2
Baseline results.

	(1)	(2)	(3)	(4)	(5)
Panel A. IV Estimation: First-stage (Dependent Variable: FDI_Intensity)					
FDI _{lib} × POST02 _t	0.0218*** (0.0067)	0.0211*** (0.0060)	0.0180*** (0.0059)	0.0176*** (0.0059)	0.0177*** (0.0058)
Panel B. IV Estimation: Second-stage (Dependent Variable: lnq)					
FDI_Intensity _{it}	−0.9833* (0.5567)	−1.3638** (0.6658)	−1.2654** (0.6434)	−1.2865** (0.6331)	−1.5457** (0.6483)
Output tariff _{it}				0.0757*** (0.0232)	0.0753*** (0.0233)
Input tariff _{it}				−0.0783 (0.0502)	−0.0810 (0.0501)
Panel C. Reduced-form Estimation (Dependent Variable: lnq)					
FDI _{lib} × POST02 _t	−0.0414* (0.0244)	−0.0667*** (0.0254)	−0.0683*** (0.0250)	−0.0674*** (0.0251)	−0.0677*** (0.0244)
Output tariff _{it}				0.0800*** (0.0231)	0.0804*** (0.0231)
Input tariff _{it}				−0.1184** (0.0514)	−0.1315*** (0.0507)
Firm FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
FDI determinants × year	No	Yes	Yes	Yes	Yes
SOE privatization × year	No	No	Yes	Yes	Yes
Foreign tariffs	No	No	No	Yes	Yes
Time-varying firm controls	No	No	No	No	Yes
N	403 255	403 255	403 255	403 255	402 348

Note: Panels A and B report the results of first and second-stage IV estimation, respectively. Panel C reports results of the corresponding reduced-form estimation. Our sample is the indigenous domestic firms, which is defined as firms that enter the market before 2002 with 100% domestic equity share for all the time. **FDI Determinants** include interactions of year dummies with new product intensity, export intensity, number of firms and average age of firms at CIC four-digit industry level in year 2000. **SOE Privatization** includes interactions of year dummies with share of SOE firms in the total number of firms in 2000. **Foreign Tariffs** include time-varying weighted average foreign tariffs faced by Chinese exports. **Time-varying Firm Controls** include interactions of year dummies with firm total sales and capital-labor ratio in the initial year. Robust standard errors in parentheses, clustered at CIC 4-digit industry-year level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

of Table 3. The IV estimated coefficients on both upstream and downstream FDIs are positive and significant, while those from reduced-form estimation are both positive but insignificant. The results are consistent with the literature, especially the seminal paper of Javorcik (2004), which identifies a positive productivity spillover from the upstream FDI inflows. More importantly, the horizontal effect of FDI remains negative and significant after controlling for upstream and downstream FDI.

4.2. Alternative quality measures

Different methods have been proposed to measure product quality in the literature. In our baseline regressions, we adopt the methods proposed by Khandelwal et al. (2013). To examine whether our results are driven by this specific quality measure, we use other measures in robustness checks.

Constant elasticity. In our baseline regressions, we use industry-level elasticities to estimate quality. Alternatively, we could assume constant elasticity for all industries as in Khandelwal et al. (2013) and Fan et al. (2015). According to Anderson and van Wincoop (2004), the reasonable range of the estimated elasticity is [5, 10]. We now use constant elasticity with the value of $\sigma = 5$ and $\sigma = 10$ to re-estimate the quality, denoted by $\ln q_5$ and $\ln q_{10}$, respectively. The estimation results obtained using these two quality measures are presented in columns (2) and (3) of Table 3. We find that the IV and the reduced-form estimates are all negative and significant, which are consistent with our baseline estimates.

Unit value. Unit value, or unit price, is another commonly used quality measure for its easy accessibility, simplicity, and comparability. Although this measure has many shortcomings, such as the difficulty of distinguishing quality from markup or firms' pricing strategy, Feenstra and Romalis (2014) find that "the observed differences in export unit-values are attributed predominantly to quality." As a robustness check, we use unit value to proxy product quality, denoted by $\ln p$. Specifically, we follow De Loecker et al. (2016) and Fan et al. (2015) in computing unit value: deflated export value divided by physical quantity of the

exported products. Instead of including firm and year fixed effects, we include firm, product, and destination-year fixed effects to be consistent with the estimated quality measure.¹⁵ We present the regression results obtained from using unit value as export quality in column (4) of Table 3, which shows consistency with our baseline results.

Relative unit value. The unit value measure does not allow cross-product quality comparison. Manova and Yu (2017) introduce the relative log price for each product-destination-year cell to standardize the quality measure so that we can make cross-product comparisons. As illustrated by Manova and Yu (2017), we consider a firm that exports printers and cellphones. The quality of the printers exported by the firm is q_p times better than the average printers in the market, and the quality of the cellphones exported by the firm is q_c times better than the average cellphones in the market. Then, we can use q_p and q_c as the measured relative quality levels of the two products, respectively. If $q_p > q_c$, the firm can be considered better at producing high-quality printers than cellphones. Following this idea, we let $\ln p_{fhet}$ denote the log price of product h exported to destination c by firm f in year t , and $\overline{\ln p_{het}}$ denote the average log price of product h exported by all Chinese firms to destination c in year t . Then, firm f 's relative unit value of product h exported to destination c in year t is defined by $R \ln p_{fhet} = \ln p_{fhet} - \overline{\ln p_{het}}$. We report results obtained with this quality measure in column (5) of Table 3. Our key estimates are still significant and negative both in the IV and reduced-form estimations.

4.3. Alternative measures of FDI intensity

In our baseline regressions, we use firm's output as the weights to construct FDI intensity (see formula (2)). This measure has been used

¹⁵ The product and destination-year fixed effects have been implicitly controlled when using $\ln q_5$, $\ln q_{10}$, and $\ln q$, as these fixed effects have been included in the quality estimation procedures.

Table 3
Robustness.

	Vertical linkages	Different quality measures				Different FDI measures		Destination switching
	(1) lnq	(2) lnq5	(3) lnq10	(4) lnq	(5) rlnq	(6) lnq	(7) lnq	(8) DestGDPP
Panel A. IV Estimation: Second-stage								
FDI_Intensity _{it}	−1.5546*** (0.5642)	−1.6348*** (0.5114)	−1.5637*** (0.4723)	−1.0165*** (0.3677)	−1.0670*** (0.3266)			−0.6049 (1.3257)
FDI_Intensity_up _{it}	0.2004** (0.0893)							
FDI_Intensity_down _{it}	0.0372** (0.0183)							
FDI_Intensity_Asset _{it}						−1.6369*** (0.5823)		
FDI_Intensity_Labor _{it}							−3.3229** (1.3740)	
Panel B. Reduced-form Estimation:								
FDI_lib _i × POST02 _t	−0.0778*** (0.0260)	−0.0781*** (0.0168)	−0.0683*** (0.0153)	−0.0442*** (0.0137)	−0.0411*** (0.0113)	−0.0677*** (0.0244)	−0.0677*** (0.0244)	−0.0070 (0.0332)
∑ _{j≠i} FDI_lib _j × a _{ji} × POST02 _t	0.0342 (0.0407)							
∑ _{j≠i} a _{ij} × FDI_lib _j × POST02 _t	0.0050 (0.0109)							
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	No	No	Yes	Yes	Yes
Product FE	No	No	No	Yes	Yes	No	No	No
Destination × year FE	No	No	No	Yes	Yes	No	No	No
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	402 348	409 169	409 169	408 839	408 839	402 348	402 348	22 204

Note: Controls include all control variables as in column (5) of Table 2. Robust standard errors in parentheses, clustered at CIC 4-digit industry-year level.

* p < 0.10, ** p < 0.05, *** p < 0.01.

by many researchers (e.g., Javorcik, 2004; Lu et al., 2017) to estimate FDI spillovers. However, other researchers have used alternative weights in their construction of FDI intensity. Barrios et al. (2011) find that measures with different weights might lead to different results. To address the concern that our findings may be driven by the specific measure we use, we now use firm's total asset and labor, respectively, as the weights to construct alternative FDI intensity indexes for robustness checks.

Columns (6) and (7) of Table 3 display the regression results with FDI intensity weighted by total asset and labor, respectively. The negative and significant effect of FDI on export quality still holds and is quantitatively similar to the one obtained with FDI intensity weighted by output. Thus, our main finding is robust to alternative FDI intensity measures.

4.4. Market switching

Suppose that a firm switches the export destination of its product. Then, by our construction of product quality measure, the quality of the same product is likely to change. For example, suppose that a Chinese firm exports bicycles to Mexico before and now switches to the US market. Also suppose that competition in the Mexican market is less intensive than that in the US market because there are more high-quality bicycles in the US market. Then, the calculated quality of the same bicycles is reduced by construction.¹⁶ We expect that competition in high-income countries are different from (maybe more intensive than) that in low-income countries. To see whether this is a possible reason for the negative effects of FDI on Chinese firms' export quality, we examine whether FDI induces Chinese firms to systematically switch their product destinations from low-income to high-income destinations or the other way around.

Specifically, we construct a measure of average destination income, denoted by $DestGDPP_{ft}$, which is the average GDP per capita of firm f 's export destinations in year t : $DestGDPP_{ft} = \ln(\sum_c s_{fct} * GDPP_c)$,

where $GDPP_c$ refers to the GDP per capita in 2000 (the initial year of the sample period) of country c and s_{fct} is the value share of firm f 's exports to country c in year t . We replace the outcome variable in model (3) with $DestGDPP_{ft}$ (the average income of firm f 's export destinations in year t). Column (8) of Table 3 presents the regression results, which show that firms in the reformed industries do not have significantly different average destination GDP per capita from those in the unreformed industries. This indicates that our main result is not driven by the change in measured quality due to market switching

5. A model of product quality, FDI and export

The objective of this section is two-fold. First, we introduce a simple single-product model to analyze the impact of foreign entry on domestic firms' product quality. This theoretical framework helps us understand the basic mechanism behind the empirical finding obtained in the preceding section. Second, we construct a multiproduct model which guides us to explore other margins in which FDI affects product quality.

5.1. A single-product model

Consider a two-country model with China ($i = 0$) and Foreign ($i = 1$). The preference of the representative consumer in each market is characterized by a constant elasticity of substitution (CES) utility function of horizontally and vertically differentiated products as follows:

$$U_i = \left(\int_{\omega \in \Omega_i} (q_i(\omega) x_i(\omega))^{\frac{\sigma-1}{\sigma}} d\omega \right)^{\frac{\sigma}{\sigma-1}}, \quad \sigma > 1, \quad (4)$$

where ω is the index of product, Ω_i is the set of products available in market i , $x_i(\omega)$ is the quantity of the product consumed, $q_i(\omega)$ is the quality of the product, and σ is the elasticity of substitution between products.

Utility maximization subject to budget constraint leads to the following demand function:

$$x_i(\omega) = \Phi_i q_i(\omega)^{\sigma-1} p_i(\omega)^{-\sigma}, \quad \text{where } \Phi_i \equiv \frac{E_i}{\int_{\omega \in \Omega_i} \left(\frac{q(\omega)}{p(\omega)} \right)^{\sigma-1} d\omega}, \quad (5)$$

¹⁶ For example, Xiaomi, a Chinese mobile phone, may be perceived as low quality in the US but high quality in India.

$p_i(\omega)$ is the price, and E_i is the exogenous income country i spends on these products.

Turning to the supply side, we consider Chinese firms first. As in Melitz (2003), each firm needs to incur a sunk cost, e , to enter the industry. Upon entry, each firm draws a productivity level φ from a Pareto distribution with c.d.f. $G(\varphi) = 1 - (\bar{\varphi}/\varphi)^k$, with $\varphi > \bar{\varphi} > 0$, to produce its (single) product. Following Kugler and Verhoogen (2012), we assume that each firm needs to incur a fixed cost to obtain a quality. This fixed cost, denoted by $f(q)$, captures the overhead costs, such as R&D expenditures. Specifically, we assume the following fixed cost of quality: $f(q) = \frac{1}{\beta} q^\beta$, where parameter β is the elasticity of cost increase to the change in quality. To produce one unit of its product, a firm with its drawn productivity φ incurs a cost $\frac{1}{\varphi}$, which is independent of the quality level. Since our empirical result is not about Chinese firms' export decision and sorting, we assume that the Chinese firms do not incur variable costs of export (such as tariff and transport cost) and fixed cost of entering the foreign market (such as market development) to simplify the analysis. Hence, if a Chinese firm enters the domestic market, it also exports to the foreign market. At the end of this subsection, we discuss the equilibrium results when this assumption is relaxed.

We now turn to the foreign firms. Each foreign firm incurs a sunk cost e_F to enter the industry. Upon entry, each firm draws a productivity level φ_F from a Pareto distribution with c.d.f. $G_F(\varphi_F) = 1 - (\bar{\varphi}_F/\varphi_F)^{k_F}$, with $\varphi_F > \bar{\varphi}_F > 0$, to produce its product. Given that our focus is on the Chinese firms' quality choice, we simplify the model regarding the foreign firms' quality choice, assuming that the foreign firms have identical product quality, denoted by q_F and incur no cost of obtaining the quality. The foreign firms can access the Chinese market either through exporting or FDI. To export to China, a foreign firm incurs a fixed export cost f_X and an iceberg variable trade cost $\tau > 1$. To enter China via FDI, a foreign firm incurs a fixed investment cost f_I .

Next, we describe the markets. Each market is characterized by monopolistic competition. Following the literature, we assume that FDI has technology spillovers over the Chinese firms. We also allow for technology spillovers from foreign firms' export to China although it is commonly accepted that technology spillover from exports, if it exists, is lower than that from FDI. Specifically, assume that a Chinese firm's drawn productivity φ becomes $s\varphi$, where s denotes the degree of technology spillover and is an increasing function of the mass of FDI and that of foreign exporters, denoted by M_I and M_X , respectively. Thus, $s = s(M_I, M_X)$ with $s(0, 0) = 1$ (i.e., in the absence of foreign entry), $\frac{\partial s}{\partial M_I} \geq 0$, and $\frac{\partial s}{\partial M_X} \geq 0$.

We now analyze the model. Denote the sum of the two markets by $\Phi = \Phi_0 + \Phi_1$.

First, a Chinese firm with the drawn productivity φ has the following profit:

$$\pi(q, p_0, p_1; \varphi) = \sum_{i \in \{0,1\}} \left(p_i - \frac{1}{s\varphi} \right) x_i - \frac{1}{\beta} q^\beta - e, \quad \text{s.t. } x_i = \Phi_i q^{\sigma-1} p_i^{-\sigma}. \quad (6)$$

If the firm decides to stay in production, it chooses $\{q, p_0, p_1\}$ to maximize the above profit. The first-order conditions yield the following optimal price and quality:

$$p_i(\varphi) = \frac{1}{\rho \varphi s} \quad \text{and} \quad q(\varphi) = [\rho^\sigma (\varphi s)^{\sigma-1} \Phi]^\frac{1}{\theta}, \quad \text{where } \rho \equiv \frac{\sigma-1}{\sigma} \quad \text{and} \quad \theta \equiv 1 - \sigma + \beta. \quad (7)$$

The firm's markup is $\frac{1}{\rho} = \frac{\sigma}{\sigma-1}$. Assume $\theta > 0$.

Evidently from (7), firms produce higher quality when the total demand (Φ) is larger, productivity (φ) is higher, and technology spillover (s) is stronger. Importantly, foreign entry (FDI and/or export) affects Chinese firms' product quality through its influence on s , which is

called the *spillover effect*, and Φ , which is called the *competition effect*. While the spillover effect is positive, the competition effect is negative as foreign entry intensifies competition which lowers the equilibrium aggregate demand Φ .

Using (7) in (6), we obtain the optimal profit:

$$\pi(\varphi) = \frac{\theta C}{\sigma \beta} \Phi^\frac{\theta}{\sigma} (s\varphi)^{\beta\eta} - e, \quad \text{where } \eta \equiv \frac{\sigma-1}{\theta} \quad \text{and} \quad C \equiv \rho^\frac{(\sigma-1)(\beta+1)}{\theta}.$$

Profit increases with productivity φ . Let φ^* be the productivity cutoff of the Chinese firms for staying in production, which is determined by the zero-profit condition: $\pi(\varphi^*) = 0$. Firms with $\varphi \geq \varphi^*$ stay in production, and others exit the industry. Solving the zero-profit condition yields the following:

$$\varphi^* = \left(\frac{\sigma \beta e}{\theta C} \right)^\frac{1}{\beta\eta} \frac{1}{s} \Phi^\frac{1}{1-\sigma}. \quad (8)$$

Next, we consider foreign firms' decisions. If a foreign firm operates in its local market, it chooses price to maximize the following profit it can derive from the local market:

$$\pi_F(p_F; \varphi_F) = (p_F - \frac{1}{\varphi_F}) x_F - e_F, \quad \text{s.t. } x_F = \Phi_1 q_F^{\sigma-1} p_F^{-\sigma}.$$

Facing a given market condition Φ_0 in China, if a foreign firm exports to China, its export profit is

$$\pi_X(p_X; \varphi_F) = (p_X - \frac{\tau}{\varphi_F}) x_X - f_X, \quad \text{s.t. } x_X = \Phi_0 q_F^{\sigma-1} p_X^{-\sigma};$$

if it chooses FDI to enter China, its FDI profit is

$$\pi_I(p_I; \varphi_F) = (p_I - \frac{1}{\varphi_F}) x_I - f_I, \quad \text{s.t. } x_I = \Phi_0 q_F^{\sigma-1} p_I^{-\sigma},$$

where subscript X denotes export and I denotes FDI. Due to constant markup pricing, that is, $p_F = p_I = \frac{1}{\rho \varphi_F}$ and $p_X = \frac{\tau}{\rho \varphi_F}$, we obtain the foreign firm's profits from local market, exporting, and FDI, respectively, as

$$\pi_F(\varphi_F) = B (\varphi_F)^{\sigma-1} \Phi_1 - e_F, \quad \pi_X(\varphi_F) = B \left(\frac{\varphi_F}{\tau} \right)^{\sigma-1} \Phi_0 - f_X,$$

$$\text{and } \pi_I(\varphi_F) = B (\varphi_F)^{\sigma-1} \Phi_0 - f_I,$$

where $B \equiv (1 - \rho)(\rho q_F)^{\sigma-1}$.

We follow the literature to assume $\tau^{\sigma-1} f_X < f_I$. In addition, we assume that the foreign market is sufficiently large. Then, the productivity cutoffs of entering the local market, exporting, and FDI are given, respectively, as

$$\varphi_F^* = \left(\frac{e_F}{B \Phi_1} \right)^\frac{1}{\sigma-1}, \quad \varphi_X = \tau \left(\frac{f_X}{B \Phi_0} \right)^\frac{1}{\sigma-1}, \quad \text{and } \varphi_I = \left[\frac{f_I - f_X}{B \Phi_0 (1 - \tau^{1-\sigma})} \right]^\frac{1}{\sigma-1}, \quad (9)$$

where $\varphi_F^* < \varphi_X < \varphi_I$. That is, foreign firms with productivity higher than φ_F^* enter the local market, those with productivity within the range $[\varphi_X, \varphi_I]$ enter the Chinese market by exporting, and those with productivity higher than φ_I enter the Chinese market by FDI.

Let Ω_H , Ω_F , Ω_X , and Ω_I be the sets of products produced by the Chinese firms, foreign firms for their local market, foreign exporters, and the FDI firms, respectively. Then, from (5), we obtain

$$\Phi_0 = \frac{E_0}{\int_{\omega \in \Omega_H} \left(\frac{q(\omega)}{\rho(\omega)} \right)^{\sigma-1} d\omega + \int_{\omega \in \Omega_X} \left(\frac{q_F}{\rho_X} \right)^{\sigma-1} d\omega + \int_{\omega \in \Omega_I} \left(\frac{q_F}{\rho_I} \right)^{\sigma-1} d\omega} \quad (10)$$

and

$$\Phi_1 = \frac{E_1}{\int_{\omega \in \Omega_H} \left(\frac{q(\omega)}{\rho(\omega)} \right)^{\sigma-1} d\omega + \int_{\omega \in \Omega_F} \left(\frac{q_F}{\rho_F} \right)^{\sigma-1} d\omega} \quad (11)$$

To see how the quality of Chinese firms changes in response to the FDI cost, f_I , based on the optimal quality function (7), we differentiate $\log(q(\varphi)^\theta)$ with respect to f_I and obtain:

$$\frac{\partial \log(q(\varphi)^\theta)}{\partial f_I} = (\sigma - 1) \frac{1}{s} \left(\frac{\partial s}{\partial M_I} \frac{\partial M_I}{\partial f_I} + \frac{\partial s}{\partial M_X} \frac{\partial M_X}{\partial f_I} \right) + \frac{1}{\Phi} \frac{\partial \Phi}{\partial f_I}. \quad (12)$$

Hence, the first term of (12) captures how FDI liberalization (a reduction in f_I) affects the positive technology spillovers of foreign entry on Chinese firms' product quality. On the one hand, from (9), a smaller f_I lowers φ_I , which enlarges the FDI set Ω_I , i.e., $\frac{\partial M_I}{\partial f_I} < 0$, where M_I is the mass of Ω_I . Consequently, the increase in FDI generates more spillovers. On the other hand, from (9), a smaller f_I reduces the set of exporters because $\Omega_X = [\varphi_X, \varphi_I]$ and it leaves φ_X unchanged while reducing φ_I , that is, $\frac{\partial M_X}{\partial f_I} > 0$, where M_X is the mass of Ω_X . Thus, FDI liberalization reduces the spillover from export by reducing the number of exporters. Therefore, the net impact of FDI liberalization on spillover depends on the relative degrees of the spillovers from export to FDI due to certain exporters switching to FDI after the FDI liberalization. Under the common assumption that FDI has a stronger spillover than exporting, we expect to see the net increase in the spillover effect from FDI liberalization, in which case, the first term of (12) is negative.

The above analysis allows us to obtain the first two inequalities below and we prove the last inequality in the Appendix:

$$\frac{\partial M_I}{\partial f_I} < 0, \quad \frac{\partial M_X}{\partial f_I} > 0, \quad \text{and} \quad \frac{\partial \Phi}{\partial f_I} > 0. \quad (13)$$

The last inequality of (13), which indicates that the second term of (12) is positive, suggests that FDI liberalization makes the competition effect stronger as it reduces the aggregate demand Φ . As argued above, FDI liberalization induces some exporters to switch to FDI. Although the policy does not change the set of foreign firms entering the Chinese market, the switch intensifies competition in the Chinese market because FDI firms have lower prices ($\frac{1}{\rho\varphi_F}$) than exporters ($\frac{\tau}{\rho\varphi_F}$), making Φ_0 smaller ($\frac{\partial \Phi_0}{\partial f_I} > 0$). Consequently, the Chinese firms reduce their investment in product quality. This in turn reduces the competition in the foreign market as all Chinese firms compete in the foreign market with lower product quality. The reduced competition results in a larger Φ_1 , which in turn raises the Chinese firms' incentive to invest in product quality. However, this feedback effect cannot dominate the effect from the domestic market (Φ_0). As a result, the Chinese firms reduce their product quality in equilibrium.

The above analysis allows us to conclude about the impact of FDI liberalization on Chinese firms' product quality as follows. *FDI liberalization policy increases both the positive technology spillover effect and the negative competition effect. If the increase in the spillover effect is relatively small, then the negative market competition effect dominates and therefore the Chinese firms reduce their product quality; and vice versa.*

Our empirical finding in Section 3 that FDI liberalization reduces Chinese firms' product quality suggests that the negative market competition dominates the positive spillover effect in China.

Extension. We now consider the case where there exists fixed (f_0) as well as variable costs ($\tau_0 > 1$) to the Chinese firms that export their products to the foreign market. In the Appendix, we show that there exists an $\tilde{\varphi}^*$ ($> \varphi^*$) such that Chinese firms with productivity drawn higher than this cutoff stay in the industry. There exists an $\tilde{\varphi}_X^*$ ($> \tilde{\varphi}^*$) such that Chinese firms with productivity drawn higher than this cutoff export their products to the foreign market. A Chinese firm's optimal quality decision is given below:

$$q(\varphi) = [\rho^\sigma (\varphi s)^{\sigma-1} (\Phi_0 + v\tau_0^{1-\sigma} \Phi_1)]^{\frac{1}{\theta}}, \quad (14)$$

where $v = 1$ if it is an exporter and $v = 0$ otherwise. It is clear that non-exporters have lower product quality, not only because they are less productive, but also because their quality investment has lower

marginal benefits as they do not have the export sales ($v = 0$). Differentiating quality with respect to FDI cost yields:

$$\begin{aligned} \frac{\partial \log(q(\varphi)^\theta)}{\partial f_I} &= (\sigma - 1) \frac{1}{s} \left(\frac{\partial s}{\partial M_I} \frac{\partial M_I}{\partial f_I} + \frac{\partial s}{\partial M_X} \frac{\partial M_X}{\partial f_I} \right) \\ &\quad + \frac{1}{\Phi_0 + v\tau_0^{1-\sigma} \Phi_1} \left(\frac{\partial \Phi_0}{\partial f_I} + v\tau_0^{1-\sigma} \frac{\partial \Phi_1}{\partial f_I} \right). \end{aligned}$$

The above expression is similar to (12). The first term is identical to the first term of (12), which does not depend on the exporter index v . It indicates that the positive spillover effect of foreign entry on both exporters and non-exporters are the same, which is clear from the definition of the s function. There is an additional term for exporters ($v = 1$) compared to non-exporters ($v = 0$) in the second term, which represents the negative competition effect. This additional term softens the competition effect as the aggregate demand in the foreign market increases. *Hence, under the condition that competition effect dominates spillover effect, FDI liberalization reduces Chinese non-exporters' product quality more than that of the exporters.*

5.2. A multiproduct model

While the single-product model provides the economics for the impact of FDI on Chinese firms' product quality, in this subsection, we extend it to a multiproduct setting to investigate more details of the mechanism on how firms adjust quality along different product margins within firms, which in turn guides further empirical analysis in the next section. Based on the analysis and results from the preceding subsection, we can drop the export choice of the foreign firms to simplify the model and analysis. In addition, we assume single product for each foreign firm as we are not to analyze foreign firms' product adding and dropping. In what follows, we do not repeat the features of the single-product model which are kept in the multiproduct model, and just describe the key extensions of the model.

Following Bernard et al. (2010), we introduce exogenous consumer tastes for each product/variety in the Chinese market. In particular, we revise the Chinese consumers' utility function as

$$U_0 = \left(\int_{\omega \in \Omega_0} (q_0(\omega) \lambda_0(\omega) x_0(\omega))^{\frac{\sigma-1}{\sigma}} d\omega \right)^{\frac{\sigma}{\sigma-1}}, \quad \sigma > 1,$$

where $\lambda_0(\omega)$ is the subjective consumer taste on product ω , while $q_0(\omega)$ is the quality of the product which is determined by the firm producing that product. In contrast, to simplify the analysis without affecting the intuition and results in a qualitative way, we assume that the foreign country's utility function is the same as that in (4) with $i = 1$, that is, there are no subjective consumer tastes across products.

Each Chinese firm incurs a fixed sunk cost e to enter the industry. Upon entry, in addition to the productivity drawn φ , a firm draws a capability of multiproduct management, m , from a distribution $G_M(m)$, where m is the maximum number of products the firm can produce/manage.¹⁷ The firm has the same productivity in all of its products. In Bernard et al. (2010), consumer tastes are assumed as a continuous and exogenous variable $\lambda(\omega)$. We assume that the consumer tastes (in the Chinese market) shocks are discrete, high or low, in order to focus on the quality decision of the firms. In particular, we assume that $\lambda_0(\omega)$ is either equal to 1 (a positive shock) or 0 (a negative shock).

Then, the firm decides whether to stay or exit the industry. If it exits, it disappears from the industry. If it stays, it makes an investment on the quality of each of its products, facing the investment cost $f(q) = \frac{1}{\beta} q^\beta$. Since demands for all positive-shock products are symmetric and a firm has the same productivity (marginal cost) applied to all its products, its decisions/outcomes (price, output, and quality, and profit) on all its

¹⁷ Qiu and Zhou (2013) analyze a model in which a firm's product scope is endogenized.

products are the same.¹⁸ Suppose that there are many positive shocks of consumer tastes such that if it is profitable for a firm to produce one product, it is also profitable for the firm with the management capability drawn m to produce m products.¹⁹

Then, if a Chinese firm with productivity φ and management capability m stays in the industry, it has the following profit function:

$$\pi(\varphi, m) = \sum_{i \in \{0,1\}} \sum_{\omega=1}^m \left(p_i(\omega) - \frac{1}{s\varphi} \right) x_i(\omega) - \frac{1}{\beta} \sum_{\omega=1}^m q(\omega)^\beta - e.$$

The firm chooses the quality of each product $q(\omega)$, and the price of each product to maximize the above profit. Based on the analysis of the single-product model (especially (7)), we obtain the optimal profit as

$$\pi(\varphi, m) = \frac{m\theta C}{\sigma\beta} \Phi^{\frac{\theta}{\sigma}} (s\varphi)^{\beta\eta} - e.$$

A firm will stay in the market if and only if the following inequality holds:

$$m\varphi^{\beta\eta} \geq \frac{\sigma\beta e}{\theta C} \frac{1}{s^{\beta\eta}} \Phi^{\frac{\theta}{\sigma}}.$$

That is, firms with sufficiently high productivity and management capability will stay in the market.

Each FDI firm has the following profit from the Chinese market: $\pi_F(\varphi_F) = (p_{F1} - \frac{1}{\varphi_F})x_F - f_I$, where $x_F = \Phi_0 q_F^{\sigma-1} p_F^{-\sigma}$. With the optimal pricing $p_F = \frac{1}{\rho s \varphi_F}$, we obtain its optimal profit as:

$$\pi_F(\varphi_F) = (1 - \rho)(\rho s q_F)^{\sigma-1} (\varphi_F)^{\sigma-1} \Phi_0 - f_I. \quad (15)$$

$\pi_F(\varphi_F)$ is increasing in φ_F . The productivity cutoff for FDI entry φ_I is determined by $\pi_F(\varphi_I) = 0$, which is $\varphi_I = \tau \left(\frac{f_I}{B\Phi_0} \right)^{\frac{1}{\sigma-1}}$, and $\frac{\partial \varphi_I}{\partial f_I} > 0$.

Let Ω_H , Ω_F , and Ω_I be the sets of products produced by the Chinese firms, foreign firms for their local market, and the FDI firms, respectively. Then, from (5), we obtain

$$\Phi_0 = \frac{E_0}{\int_{\omega \in \Omega_H} \left(\frac{q(\omega)}{p(\omega)} \right)^{\sigma-1} d\omega + \int_{\omega \in \Omega_I} \left(\frac{q_F}{p_I} \right)^{\sigma-1} d\omega}$$

and Φ_1 which is the same as (11) in expression.

Results obtained in the above analysis jointly determine the equilibrium.

Suppose that (i) there are changes in consumer tastes in the Chinese market, and (ii) there is an FDI liberalization, captured by a drop of f_I . Then, if a firm receives a negative consumer taste shock in some of its products, it drops those products as the demands become zero ($\lambda_0(\omega) = 0$). It adds new products which have positive consumer taste shocks, keeping the same number of products (m) (by assumption). The question is how it invests in the quality of those newly added products. It is clear that that qualities of those new products are determined in the new equilibrium, with a smaller f_I . Let $\tilde{q}(\omega)$ be the quality of the newly added product produced by the firm that has this variety, Ω_{HO} be the set of products that the Chinese firms keep producing, and Ω_{HN} be the set of products that the Chinese firms add after the shocks. Then, we have

$$\Phi_0 = \frac{E_0}{\int_{\omega \in \Omega_{HO}} \left(\frac{q(\omega)}{p(\omega)} \right)^{\sigma-1} d\omega + \int_{\omega \in \Omega_{HN}} \left(\frac{\tilde{q}(\omega)}{p(\omega)} \right)^{\sigma-1} d\omega + \int_{\omega \in \Omega_I} \left(\frac{q_F}{p_I} \right)^{\sigma-1} d\omega},$$

¹⁸ If the positive consumer taste shocks vary in degree (as in Bernard et al., 2010), the qualities of a firm's products are also different.

¹⁹ Bernard et al. (2010) and Qiu and Zhou (2013) analyze whether high productivity firms produce more or fewer products. We stay away from this issue to focus on product quality.

and

$$\Phi_1 = \frac{E_1}{\int_{\omega \in \Omega_{HO}} \left(\frac{q(\omega)}{p(\omega)} \right)^{\sigma-1} d\omega + \int_{\omega \in \Omega_{HN}} \left(\frac{\tilde{q}(\omega)}{p(\omega)} \right)^{\sigma-1} d\omega + \int_{\omega \in \Omega_F} \left(\frac{q_F}{p_F} \right)^{\sigma-1} d\omega}.$$

While the qualities of the existing products ($q(\omega)$) do not change, FDI liberalization affects those of the newly added products ($\tilde{q}(\omega)$), and similar to (12), we obtain:

$$\frac{\partial \log(\tilde{q}(\omega)^\theta)}{\partial f_I} = (\sigma - 1) \frac{1}{s} \frac{\partial s}{\partial M_I} \frac{\partial M_I}{\partial f_I} + \frac{1}{\Phi} \frac{\partial \Phi}{\partial f_I}. \quad (16)$$

There exist two opposing effects. On the one hand, the first term of (16) captures how FDI liberalization (a reduction in f_I) affects the positive technology spillovers of foreign entry on Chinese firms' product quality: $\frac{\partial M_I}{\partial f_I} < 0$, which comes from $\frac{\partial \varphi_I}{\partial f_I} > 0$. On the other hand, following the same analysis in Appendix A.2, we can prove $\frac{\partial \Phi}{\partial f_I} > 0$, which captures the negative competition effect.²⁰ Hence, as in the single-product case, we obtain that if the increase in the spillover effect is relatively small, then the negative market competition effect dominates and therefore the Chinese firms' newly added products have lower quality.

In summary, our multiproduct model predicts the following when firms face changes in consumer tastes and FDI liberalization:

- (i). Firms introduce new products to replace dropped products, keeping the product scope more or less unchanged.
- (ii). The dropped products are the average products in terms of product quality, which is an assumption of the model.
- (iii). Before FDI liberalization, the newly introduced products have on average the same quality level as continuing products. However, after the FDI liberalization, the qualities of the newly introduced products are lower than the average. This is the key prediction.

6. Empirical exploration of the mechanisms

In the preceding section, the static model provides the economic intuition for the impact of FDI on technology spillovers, domestic market competition, and product quality while the dynamic model discusses the margins of the Chinese firms' decisions in response to the increased FDI. In this section, we explore the validity of those mechanisms empirically.

6.1. Domestic market competition

We first examine whether FDI reduces Chinese firms' sales in the domestic and international markets. To this end, we introduce the following firm-level test on whether the changes in the Chinese firms' total, domestic, and export sales in the reformed industries are significantly different from those in the unreformed industries after 2002:

$$y_{fit} = \beta FDI_lib_i \times POST02_t + X'_{fit} \gamma + \lambda_f + \lambda_t + \epsilon_{ft}, \quad (17)$$

where y_{fit} is the firm-level outcome variable of interest, including total, domestic, and export sales in year t . The regression results are reported in columns (1)-(3) of Table 4. In column (1), we find that FDI has a negative and significant effect on firms' total sales. Columns (2) and (3) show that this negative effect is mainly driven by the decrease of firms' domestic sales, while the total export is not significantly affected. These results indicate that the negative competition effect mainly concentrates on the domestic market, consistent with our model prediction.

The aforementioned finding allows us to focus on the domestic market to analyze the FDI effects on export quality. The finding implies

²⁰ The only difference in the two proofs is that the qualities of a subset of the products are fixed now. To save space, we do not include this proof, but which is available upon request.

Table 4
Domestic market competition.

	Market sales			Quality			
	(1) Total sales	(2) Domestic sales	(3) Export sales	(4) Low-domestic-sales group	(5) High-domestic-sales group	(6) Interaction	(7) Pure exporters
$FDI_lib_i \times POST02_t$	-2.9786** (1.4175)	-2.9802** (1.3843)	-0.0063 (0.1514)	-0.0199 (0.0278)	-0.1725*** (0.0532)	-0.0348 (0.0274)	0.7312 (0.4792)
$FDI_lib_i \times POST02_t \times DomSaleSh_f$						-0.1192* (0.0642)	
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	22 204	22 204	22 204	266 194	135 839	402 033	16 793

Note: Controls include all control variables as in column (5) of Table 2. Robust standard errors in parentheses, clustered at CIC 4-digit industry-year level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

that firms with a larger share of domestic sales in their total sales are affected more by FDI and thus reduce their product quality more. To explore these heterogeneous effects, we divide all Chinese firms into two groups: the first group consists of those whose shares of domestic sales are larger than the median share, and the second group consists of the rest. Accordingly, we define a dummy variable $DomSaleSh_f$, which takes a value of 1 if firm f belongs to the high-domestic-sales group and 0 if otherwise.

We run regressions based on model (3) with the two groups, respectively. Column (4) of Table 4 reports the results from the low-domestic-sales group ($DomSaleSh_f = 0$), and column (5) presents the results from the high-domestic-sales group ($DomSaleSh_f = 1$). Although the FDI effects on Chinese firms' export quality are negative for both the low- and high-domestic-sales groups, the effects are significant for the latter but insignificant for the former. These findings confirm the aforementioned conjecture that FDI's negative effect on Chinese firms' export quality is through the influence on the domestic market.

We also run a triple-difference regression by multiplying domestic sales dummy $DomSaleSh_f$ to the interaction term of the FDI and reform dummies $FDI_lib_i \times POST02_t$ in model (3). We report the results in column (6) of Table 4.²¹ The estimate of the triple interaction term is negative and significant, larger than the estimate of $FDI_lib_i \times POST02_t$ in magnitude. This result further confirms that the FDI's negative impact on export quality is more pronounced for the Chinese firms that concentrate more on the domestic market.

Finally, we run a placebo test on pure exporters, which are defined as the Chinese firms that have no sales in the domestic market. The idea is that if the change in domestic market competition is the key mechanism that FDI affects Chinese firms' export quality, then we do not expect the export quality of the pure exporters to be affected. The result of the placebo test is reported in column (7) of Table 4, which confirms this conjecture, thereby reinforcing our intuition about the role of domestic market competition.

In summary, the preceding analysis provides evidence to support the prediction from the static model, that is, domestic market competition is an important channel through which FDI reduces Chinese firms' export quality.

6.2. Product churning and adding

We now explore more details on how firms respond to the change in domestic market competition. Inspired by our dynamic model in the preceding section, our analysis focuses on firms' product churning and adding.

Over a given period of time, we can classify a firm's products in each period into three types: continuing, new, and dropped products.

For our empirical analysis, we define the three types as follows. First, we identify a firm's initial set of export products. If a firm exports in the beginning year of our sample period, i.e., 2000, a firm's initial set of export products consists of all products the firm exported in 2000. If a firm did not export in 2000 but exported in 2001, the firm's initial set of exports consists of all products it exported in 2001.²² Then, in any year t , a product is defined as a *continuing product* if it is in the firm's initial set and the firm also exports it in year t ; a product is defined as a *new product* if it is not in the initial export product set and the firm exports it in year t , and a product is defined as a *dropped product* if the firm stops exporting it from year t onwards.²³ We examine the effects of FDI on the quality of these three types of products.²⁴

□ **Continuing Products.** We run regression based on model (3) using the sub-sample of continuing products. The result is reported in column (1) of Table 5, which shows that the quality difference of continuing products in the reformed industries between the pre- and post-reformed periods is not significantly different from that in the unreformed industries. That means, FDI does not have a significant impact on the quality of the continuing goods, or, the Chinese firms do not downgrade or upgrade their continuing products. This finding is consistent with the prediction of our dynamic model. The explanation from our model is that once the quality of the product has been chosen (through investment), it will not change. In reality, a firm does not change the quality of the continuing products probably because of high adjustment costs.

□ **New Products.** We introduce a dummy variable $NewProd_{fht}$, which equals to 1 if h is a new product introduced by firm f in year t , and 0 if h is a continuing product of the firm in year t . Our dynamic model predicts that the average quality of the new products introduced in the pre-reform period is similar to that of the continuing products, but the quality of the new products introduced in the post-reform period is lower than that of the continuing products due to the increased FDI competition. To test this prediction, we replace $POST_t$ by $NewProd_{fht}$ in model (3), and rerun the regressions both in the pre- and post-reform periods, respectively, excluding the dropped products.

In the above DID model, since we exclude the dropped products in the sample, the control group consists of both the new and continuing in the unreformed industries and the treatment group consists of both

²² Our data only include firms that are active in export in both the pre and post-reform periods. The pre-reform period includes 2000 and 2001.

²³ If a firm exports a product in and only in 2000, 2002, 2005, and 2006, for example, we can still consider the product as a continuing product.

²⁴ In addition to using the quality measure in our baseline model, we also use the other two alternative quality measures, i.e., the log unit price (lp) and the relative log unit price (rlp), in this product churning and adding analysis. These measures of quality are supposed to be more comparable across varieties. We find similar results using these two alternative measures, but we do not report these results to save place.

²¹ We also include other two-term interactions in the regression, but do not report the estimates in the table to save space.

Table 5
Product churning and adding.

	Quality (lnq) of					Product scope
	(1) Continuing products	New products		Dropped products		
		(2) Pre	(3) Post	(4) Pre	(5) Post	
FDI _{lib} _i × POST02 _i	−0.0099 (0.0257)					0.0322 (0.0267)
FDI _{lib} _i × New _{fht}		0.0491 (0.0923)	−0.0711* (0.0363)			
FDI _{lib} _i × Drop _{fht}				0.0078 (0.0824)	−0.0765 (0.0521)	
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
year FE	Yes	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes	Yes
N	233 019	80 035	322 421	74 533	206 050	22 204

Note: Controls include all control variables as in column (5) of Table 2. Robust standard errors in parentheses, clustered at CIC 4-digit industry-year level.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

the new and continuing products in the reformed industries. We report the regression results in columns (2) and (3) of Table 5, for the pre- and post-reform samples, respectively. We find that the results are consistent with the model prediction: in the pre-reform period, the quality of the new products on average has no significant difference with that of the continuing products, but in the post-reform period, firms in the liberalized industries tend to add new product with lower quality than their continuing product compared to that of firms in the un-liberalized industries. Thus, FDI has a significant and negative effect on the quality of the new products introduced in reformed industries after the reform.²⁵

Manova and Yu (2017) have shown that newly exported products from China tend to have lower quality than the incumbent ones. They consider the positive demand shocks from the export market. In response, the Chinese exporters add new products. The newly added peripheral products have lower quality than the core product, along the quality ladder. Our preceding empirical results add different insights to the observation of lower quality of new products. On the one hand, our study shows that the increased competition in the domestic market due to FDI is one of the reasons for the lower quality of newly exported products. On the other hand, our findings imply that if there exist certain general factors that lower the newly exported products across industries, the quality drop of the new products in the reformed industries is larger, due to the FDI effect.²⁶

□ **Dropped Products.** In our dynamic model, firms drop some products due to large and negative shocks to product-specific fixed costs. As the shocks are random, the dropped products are the average ones, and thus, the average quality of the dropped products is similar to that of the continuing products. We now examine this prediction empirically. We introduce a dummy variable $DropProd_{fht}$, which equals to 1 if h is a dropped product by firm f in year t , and 0 if h is a continuing product of firm f in year t . The new products are excluded from the sample. We replace $POST_t$ by $DropProd_{fht}$ in model (3) and rerun the regressions in the pre- and post-reform periods, respectively. The results are reported in columns (4) and (5) of Table 5, which show that the quality difference between the dropped and continuing products in

the reformed industries is not significantly different from that in the unreformed industries, in both the pre- and post-reform periods. Thus, FDI has no significant effect on the quality of the dropped products, which is significantly different from that of the continuing products.

□ **Product Scope.** In our dynamic model, firms introduce new products to replace the dropped products so as to maintain by and large the same product scope, i.e., the number of products. We now test this pattern empirically. We replace the outcome variable in model (17) with a firm's product scope in year t . The regression result is reported in column (6) of Table 5. We find that the changes in firms' product scope in the reformed industries after 2002 have no significant difference with those in the unreformed industries. Thus, FDI does not cause the Chinese firms to change their export product scope. They introduce new products just to replace the dropped products. This finding is consistent with the pattern in our dynamic model.

□ **Summary.** Our main finding from the baseline model is that FDI reduces Chinese firms' export product quality. There are many possible channels for this observed result. Our dynamic model proposes one, which is empirically confirmed by the regression results as reported in Table 5. Our proposed channel is as follows. The Chinese firms face exogenous shocks in every period which result in dropping of some existing products. Then, the firms introduce new products to replace the dropped products. After an unanticipated positive FDI shock (increased FDI), tougher competition in the domestic market reduces the Chinese firms' sales and profits, which has implications for the optimal quality of the continuing and new products. Adjusting (especially upgrading) the quality of continuing products is too costly, and thus, the firms keep the quality of continuing products unchanged. In contrast, the firms invest less in the quality of the newly introduced products because the expected sales and profit are lower than those before the FDI shock. Since the quality of the dropped products is the average quality of the firms, the resulting quality after the FDI shock is on average lower than that before the shock.

7. Conclusion

FDI has a variety of impacts on the host countries. Policy makers of the host countries are interested in the effects of FDI on the local economy in general and employment in particular. Researchers pay special attention to the spillovers effects on the domestic firms. Almost all studies in the literature of FDI spillovers focus on productivity. A few studies (e.g., Harding and Javorcik, 2012) turn to the effects of FDI on the quality of the host countries' exports. Our study focuses on FDI in China and its effects on Chinese firms' export quality. This is of particular importance because China is one of the largest FDI recipients and exporters. Our main finding is that over the period of 2000–2006, FDI reduced the average quality of Chinese firms' export products.

We obtain the aforementioned result using the IV approach. We use China's FDI policy change in 2002 as an IV to identify the causal

²⁵ Although the key coefficient in column (3) is only significant at 10%, the information it conveys is robust as the results obtained the other two alternative measures (i.e., lp and rlp) are positively significant at 1%.

²⁶ One plausible interpretation of our finding about new product with lower quality is firms' pricing strategy. Firms might set lower prices for their new products to enter foreign markets, leading to lower measured quality. However, this is not the explanation for our finding. First, we compare export quality changes across firms with different treatment intensity. Therefore, the general pricing strategy has been differenced out. Second, we find no significant effect of FDI on measured quality of new products in the pre-reform period.

impacts of FDI on export quality. The result is robust to different measures of product quality and FDI intensity, export market changes, and inclusion of vertical FDI. We provide a theoretical framework to understand the channel through which FDI exerts the impacts on Chinese firms' export quality. When the (negative) domestic market competition effect dominates the (positive) technology spillover effect, Chinese firms reduce their investment in product quality because the return to investment is lower. As the FDI policy change in 2002 is a shock to the firms, the investment impact only shows up in newly introduced products. Hence, the observed reduction in the average quality of Chinese firms' export products is due to the lower quality of the new products while the quality of the continuing products remain unchanged. Our empirical analysis also supports the above channel and predictions.

FDI is one of the most important phenomena of globalization. However, we are far from fully understanding the impacts of FDI on the host countries' economies. Although the FDI impacts on productivity have received a lot of attention from researchers, the empirical evidence is still mixed. Even for the FDI effects on export quality, our finding based on China is opposite to that found by [Harding and Javorcik \(2012\)](#) based on other developing countries. Although the empirical literature of FDI impacts is inconclusive, it is not surprising because the studies are based on data from different countries over different periods of time and use different methodologies. However, the above observation indicates that finding the signs and causal effects of FDI is not sufficient, and identifying the mechanisms of the effects becomes very important because it helps us understand the differential results in reality. Our present study makes a contribution to the literature by identifying a mechanism behind the impacts of FDI on domestic firms' export quality. More research along this direction is expected.

CRedit authorship contribution statement

Qing Liu: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Writing – original draft, Writing – review & editing. **Larry D. Qiu:** Conceptualization, Formal analysis, Methodology, Project administration, Writing – original draft, Writing – review & editing. **Chaoqun Zhan:** Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Resources, Software, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Appendix

A.1. Sample representativeness

The following table presents the comparison of firms in our working sample with those in the matched sample of the ASIFP and Custom data, and those in the ASIFP data.

A.2. Proof of the last inequality in (13)

Converting the product set to the set of firms, we obtain

$$\begin{aligned} \int_{\omega \in \Omega_F} \left(\frac{q_F}{p_F} \right)^{\sigma-1} d\omega &= \int_{\varphi_F > \varphi_F^*} (\rho \varphi_F)^{\sigma-1} dG_F(\varphi_F) \\ &= \frac{k_F \bar{\varphi}_F (\rho q_F)^{\sigma-1}}{k_F + 1 - \sigma} (\varphi_F^*)^{\sigma-1-k_F} \\ &= D e_F^{1-\frac{k_F}{\sigma-1}} \Phi_1^{\frac{k_F}{\sigma-1}-1}, \\ \text{where } D &\equiv \frac{k_F \bar{\varphi}_F (\rho q_F)^{\sigma-1} B^{\frac{k_F}{\sigma-1}-1}}{k_F + 1 - \sigma}. \end{aligned}$$

Similarly, we calculate $\int_{\omega \in \Omega_H} \left(\frac{q(\omega)}{p(\omega)} \right)^{\sigma-1} d\omega$ and use (11) to obtain

$$\Phi_1 = \frac{E_1}{\frac{k_F \bar{\varphi}_C}{k_F - \beta \eta} \left(\frac{\sigma \beta e}{\theta C} \right)^{\frac{\beta \eta - k}{\beta \eta}} s^k \Phi^{\frac{k}{\sigma-1}-1} + D e_F^{1-\frac{k_F}{\sigma-1}} \Phi_1^{\frac{k_F}{\sigma-1}-1}}. \quad (\text{A.1})$$

We further calculate $\int_{\omega \in \Omega_X} \left(\frac{q_F}{p_X} \right)^{\sigma-1} d\omega$ and $\int_{\omega \in \Omega_I} \left(\frac{q_F}{p_I} \right)^{\sigma-1} d\omega$ and use (10) to obtain

$$\Phi_0 = \frac{E_0}{\frac{k_F \bar{\varphi}_C}{k_F - \beta \eta} \left(\frac{\sigma \beta e}{\theta C} \right)^{\frac{\beta \eta - k}{\beta \eta}} s^k \Phi^{\frac{k}{\sigma-1}-1} + D F(f_I) \Phi_0^{\frac{k_F}{\sigma-1}-1}}, \quad (\text{A.2})$$

where $F(f_I) \equiv \tau^{-k_F} (f_X)^{1-\frac{k_F}{\sigma-1}} + (1 - \tau^{1-\sigma})^{\frac{k_F}{\sigma-1}} (f_I - f_X)^{1-\frac{k_F}{\sigma-1}}$.

Let $L(\Phi_0, \Phi_1) \equiv \Phi^{\frac{k}{\sigma-1}-1}$, $H \equiv \frac{k_F \bar{\varphi}_C}{k_F - \beta \eta} \left(\frac{\sigma \beta e}{\theta C} \right)^{\frac{\beta \eta - k}{\beta \eta}}$, and $J \equiv e_F^{1-\frac{k_F}{\sigma-1}}$, then (A.1) and (A.2) can be rewritten as

$$H s^k L \Phi_1 + D J \Phi_1^{\frac{k_F}{\sigma-1}-1} = E_1 \quad \text{and} \quad H s^k L \Phi_0 + D F \Phi_0^{\frac{k_F}{\sigma-1}-1} = E_0. \quad (\text{A.3})$$

Note that s , Φ_i , L , and F are all endogenous functions of f_I , while the rest are exogenous parameters.

Taking full derivatives of the two equations in (A.3) with respect to f_I , we obtain

$$\begin{aligned} H s^k \left(\frac{k}{s} \frac{\partial s}{\partial f_I} L \Phi_1 + \frac{\partial L}{\partial f_I} \Phi_1 + L \frac{\partial \Phi_1}{\partial f_I} \right) + D J \frac{k_F}{\sigma-1} \Phi_1^{\frac{k_F}{\sigma-1}-1} \frac{\partial \Phi_1}{\partial f_I} &= 0, \\ H s^k \left(\frac{k}{s} \frac{\partial s}{\partial f_I} L \Phi_0 + \frac{\partial L}{\partial f_I} \Phi_0 + L \frac{\partial \Phi_0}{\partial f_I} \right) &+ D \left(F \frac{k_F}{\sigma-1} \Phi_0^{\frac{k_F}{\sigma-1}-1} \frac{\partial \Phi_0}{\partial f_I} + \Phi_0^{\frac{k_F}{\sigma-1}-1} \frac{\partial F}{\partial f_I} \right) = 0. \end{aligned}$$

Since $\frac{\partial L}{\partial f_I} = L \frac{\frac{k}{\sigma-1}-1}{\Phi} \left(\frac{\partial \Phi_1}{\partial f_I} + \frac{\partial \Phi_0}{\partial f_I} \right)$, we obtain

$$\begin{aligned} s^k L H \left[\frac{k}{s} \frac{\partial s}{\partial f_I} \Phi_1 + \left(\frac{k}{\sigma-1} - 1 \right) \frac{\Phi_1}{\Phi} \left(\frac{\partial \Phi_1}{\partial f_I} + \frac{\partial \Phi_0}{\partial f_I} \right) + \frac{\partial \Phi_1}{\partial f_I} \right] &+ \frac{k_F}{\sigma-1} D J \Phi_1^{\frac{k_F}{\sigma-1}-1} \frac{\partial \Phi_1}{\partial f_I} = 0, \\ s^k L H \left[\frac{k}{s} \frac{\partial s}{\partial f_I} \Phi_0 + \left(\frac{k}{\sigma-1} - 1 \right) \frac{\Phi_0}{\Phi} \left(\frac{\partial \Phi_1}{\partial f_I} + \frac{\partial \Phi_0}{\partial f_I} \right) + \frac{\partial \Phi_0}{\partial f_I} \right] &+ D \left(F \frac{k_F}{\sigma-1} \Phi_0^{\frac{k_F}{\sigma-1}-1} \frac{\partial \Phi_0}{\partial f_I} + \Phi_0^{\frac{k_F}{\sigma-1}-1} \frac{\partial F}{\partial f_I} \right) = 0. \end{aligned}$$

To simplify the notations, we denote $\lambda \equiv s^k L H$ and $h_i \equiv \frac{k_F}{\sigma-1} D \Phi_i^{\frac{k_F}{\sigma-1}-1}$. Then, the above two equations can be expressed as

$$b_1 \frac{\partial \Phi_0}{\partial f_I} + c_1 \frac{\partial \Phi_1}{\partial f_I} = a_1 \quad \text{and} \quad c_0 \frac{\partial \Phi_0}{\partial f_I} + b_0 \frac{\partial \Phi_1}{\partial f_I} = a_0, \quad (\text{A.4})$$

where

$$\begin{aligned} b_i &= \lambda \left(\frac{k}{\sigma-1} - 1 \right) \frac{\Phi_i}{\Phi}, \\ a_1 &= -\lambda \frac{k}{s} \frac{\partial s}{\partial f_I} \Phi_1, \quad a_0 = -\lambda \frac{k}{s} \frac{\partial s}{\partial f_I} \Phi_0 - D \Phi_0^{\frac{k_F}{\sigma-1}-1} F', \\ c_1 &= \lambda + b_1 + J h_1, \quad c_0 = \lambda + b_0 + F h_0. \end{aligned}$$

Table A.1
Sample representativeness.

Variable	Working sample (N = 22 196)		Matched Sample (N = 220 191)			ASIFP Sample (N = 1 217 587)		
	(1) Mean	(2) S.d.	(3) Mean	(4) S.d.	(5) Dif = (1)–(3)	(6) Mean	(7) S.d.	(8) Dif = (1)–(6)
Log (sales)	11.079	1.425	10.609	1.367	0.471***	9.949	1.34	1.13***
Log (export + 1)	8.148	4.019	7.853	4.064	0.294***	2.822	4.417	5.325***
Log (Emp)	6.089	1.229	5.349	1.18	0.740***	4.814	1.115	1.275***

Note: All sample period is 2000–2006. Working Sample includes domestic manufacturing exporters which experienced the FDI shock. Matched Sample is the merged data of the ASIFP and Chinese Custom which includes the manufacturing exporters. The ASIFP sample includes all firms in ASIFP.

Then, solving (A.4) to obtain

$$\frac{\partial \Phi_0}{\partial f_I} = \frac{a_0 c_1 - a_1 b_0}{c_0 c_1 - b_0 b_1} \quad \text{and} \quad \frac{\partial \Phi_1}{\partial f_I} = \frac{a_1 c_0 - a_0 b_1}{c_0 c_1 - b_0 b_1}.$$

It is clear that $c_0 c_1 - b_0 b_1 > 0$. Let $K \equiv -\lambda \frac{k}{s} \frac{\partial s}{\partial f_I}$ and $\mu \equiv \frac{k_F}{\sigma-1} > 1$. Under the common assumption that FDI spillover is stronger than that from exporting, we have $\frac{\partial s}{\partial f_I} > 0$. Furthermore,

$$\begin{aligned} a_0 c_1 - a_1 b_0 &= (K\Phi_0 - D\Phi_0^\mu F')(\lambda + b_1 + Jh_1) - Kb_0\Phi_1 \\ &= (K\Phi_0 - D\Phi_0^\mu F')(\lambda + Jh_1) - D\Phi_0^\mu F'b_1 > 0 \\ &\quad (\because Kb_0\Phi_1 = Kb_1\Phi_0), \end{aligned}$$

$$\begin{aligned} a_1 c_0 - a_0 b_1 &= K\Phi_1(\lambda + b_0 + Fh_0) - (K\Phi_0 - D\Phi_0^\mu F')b_1 \\ &= K\Phi_1(\lambda + Fh_0) + D\Phi_0^\mu F'b_1, \end{aligned}$$

and thus,

$$\begin{aligned} a_0 c_1 - a_1 b_0 + a_1 c_0 - a_0 b_1 &= (K\Phi_0 - D\Phi_0^\mu F')(\lambda + Jh_1) \\ &\quad + K\Phi_1(\lambda + b_0 + Fh_0) > 0. \end{aligned}$$

Therefore, we have $\frac{\partial \Phi_0}{\partial f_I} > 0$ and $\frac{\partial \Phi_1}{\partial f_I} > 0$.

A.3. Impact on non-exporters

A non-exporter's profit is

$$\pi_0(q, p_0; \varphi) = \left(p_0 - \frac{1}{s\varphi}\right)x_0 - \frac{1}{\beta}q^\beta - e, \quad \text{s.t.} \quad x_0 = \Phi_0 q^{\sigma-1} p_0^{-\sigma}.$$

An exporter's profit is

$$\begin{aligned} \pi(q, p_0, p_1; \varphi) &= \sum_{i \in \{0,1\}} \left(p_i - \frac{\tau_0}{s\varphi}\right)x_i - \frac{1}{\beta}q^\beta - e - f_0, \\ \text{s.t.} \quad x_i &= \Phi_i q^{\sigma-1} p_i^{-\sigma}. \end{aligned}$$

While a firm charges $p_0(\varphi) = \frac{1}{\rho\varphi s}$ in the domestic market, it charges $p_1(\varphi) = \frac{\tau_0}{\rho\varphi s}$ in the foreign market. It allows us to write a Chinese firm's profit as, with $v = 1$ for exporters and $v = 0$ for non-exporters:

$$\pi(q; \varphi) = \frac{1}{\sigma-1} \rho^\sigma (s\varphi)^{\sigma-1} (\Phi_0 + v\tau_0^{1-\sigma}\Phi_1) q^{\sigma-1} - \frac{1}{\beta}q^\beta - e - v f_0. \quad (\text{A.5})$$

Then, maximization leads to the optimal quality as given in (14). Substituting the optimal quality to (A.5) gives the optimal profit of each firm. In particular a firm's domestic market profit, excluding the entry cost e , is

$$\pi_0(\varphi) = \frac{\theta C}{\sigma\beta} (s\varphi)^{\beta\eta} \Phi_0^{\frac{\beta}{\theta}},$$

which is increasing in φ . Thus, we obtain the entry cutoff from $\pi_E(\tilde{\varphi}^*) = e$. We have $\tilde{\varphi}^* > \varphi^*$ because $\Phi_0 < \Phi$, where φ^* is given in (8).

An exporter's profit from the foreign market, excluding the fixed cost f_0 , is

$$\pi_E(\varphi) = \frac{1}{\sigma-1} \rho^{\frac{\sigma\beta}{\theta}} (s\varphi)^{\frac{(\sigma-1)\beta}{\theta}} (\Phi_0 + v\tau_0^{1-\sigma}\Phi_1)^{\frac{\sigma-1}{\theta}},$$

which is increasing in φ . Thus, we obtain the export cutoff from $\pi_E(\tilde{\varphi}_X^*) = f_0$.

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